2 FLIP-FLOP

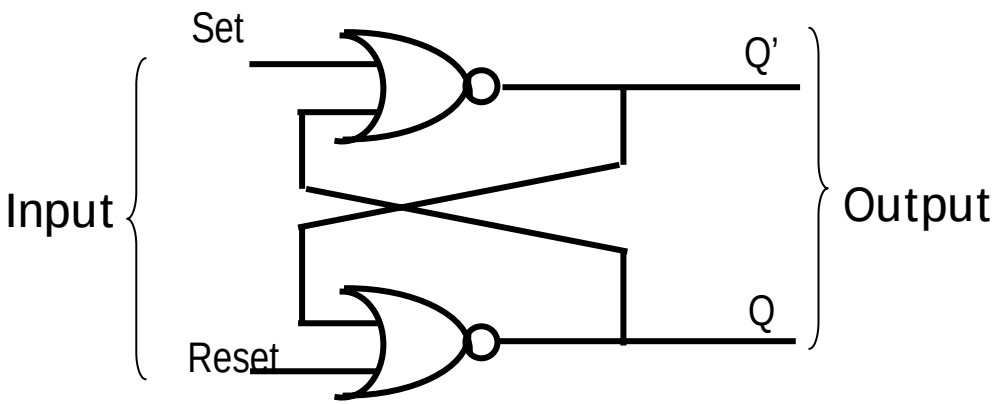
TUJUAN :

Setelah mempelajari bab ini mahasiswa diharapkan mampu :

- Menjelaskan rangkaian dasar SR-FF dan SR-FF dengan gate
- Membandingkan operasi dari rangkaian D Latch dan D-FF menggunakan timing diagram
- Menguraikan perbedaan antara *pulse-triggered* dan *edge-triggered flip-flop*
- Menjelaskan operasi rangkaian *Master Slave* JK-FF
- Membuat Toggle FF dan D-FF dari JK-FF dan SR-FF
- Menjelaskan operasi *sinkron* dan *asinkron* dari JK-FF dan D-FF menggunakan timing diagram
- Menganalisa dan mendisain rangkaian dengan Flip-flop

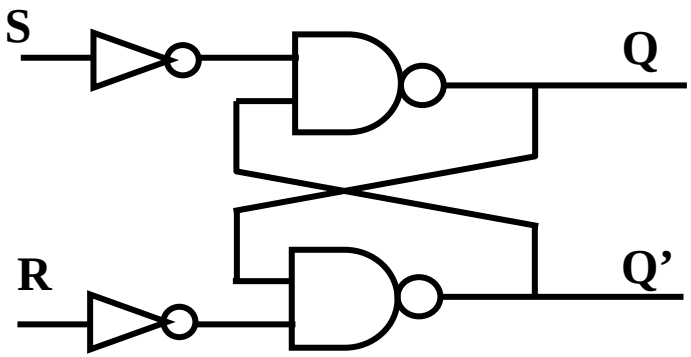
SR-FLIP-FLOP

- merupakan singkatan dari Set & Reset Flip-flop
- Dibentuk dari dua buah NAND gate atau NOR gate
- Operasinya disebut *transparent latch*, karena bagian outputnya akan merespon input dengan cara mengunci nilai input yang diberikan (*latch*) atau mengingat input tersebut.



PRESENT INPUT		PRESENT OUTPUT	NEXT OUTPUT	COMMENT
S	*R*	*Q*	*Qn*	
0	0	0	0	Hold Condition
0	0	1	1	
0	1	0	0	Flip-Flop Set
0	1	1	0	
1	0	0	1	Flip-Flop Reset
1	0	1	1	
1	1	0	*	Not Used
1	1	1	*	

Cross-NOR SR Flip-Flop



Cross-NAND SR Flip-Flop

PRESENT INPUT		PRESENT OUTPUT	NEXT OUTPUT	COMMENT
S	*R*	*Q*	*Qn*	
0	0	0	0	Hold Condition
0	0	1	1	
0	1	0	0	Flip-Flop Reset
0	1	1	0	
1	0	0	1	Flip-Flop Set
1	0	1	1	
1	1	0	*	Not Used
1	1	1	*	

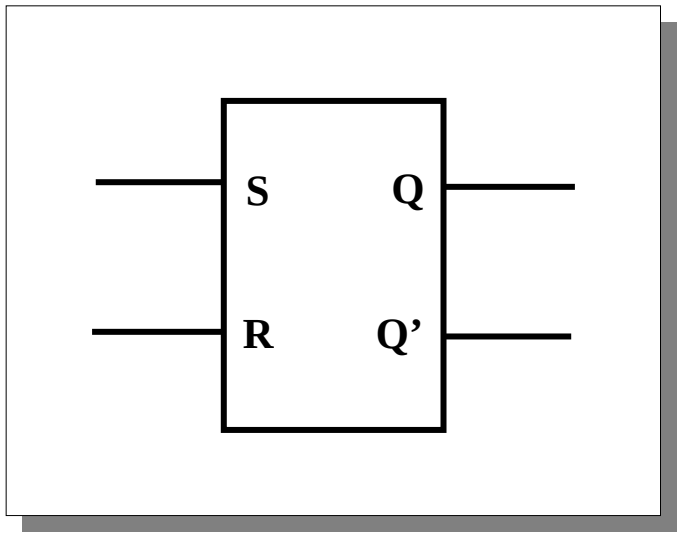
State Table dari SR-FF

Persamaan Next State SR-FF

$$Q(t+\Delta) = S(t) + \overline{R}(t)Q(t)$$

PRESENT OUTPUT	NEXT OUTPUT	NILAI EKSITASI	
Q (t)	Q (t+Δ)	S (t)	R (t)
0	0	0	d
0	1	1	0
1	0	0	1
1	1	d	0

Tabel Eksitasi dari SR-FF

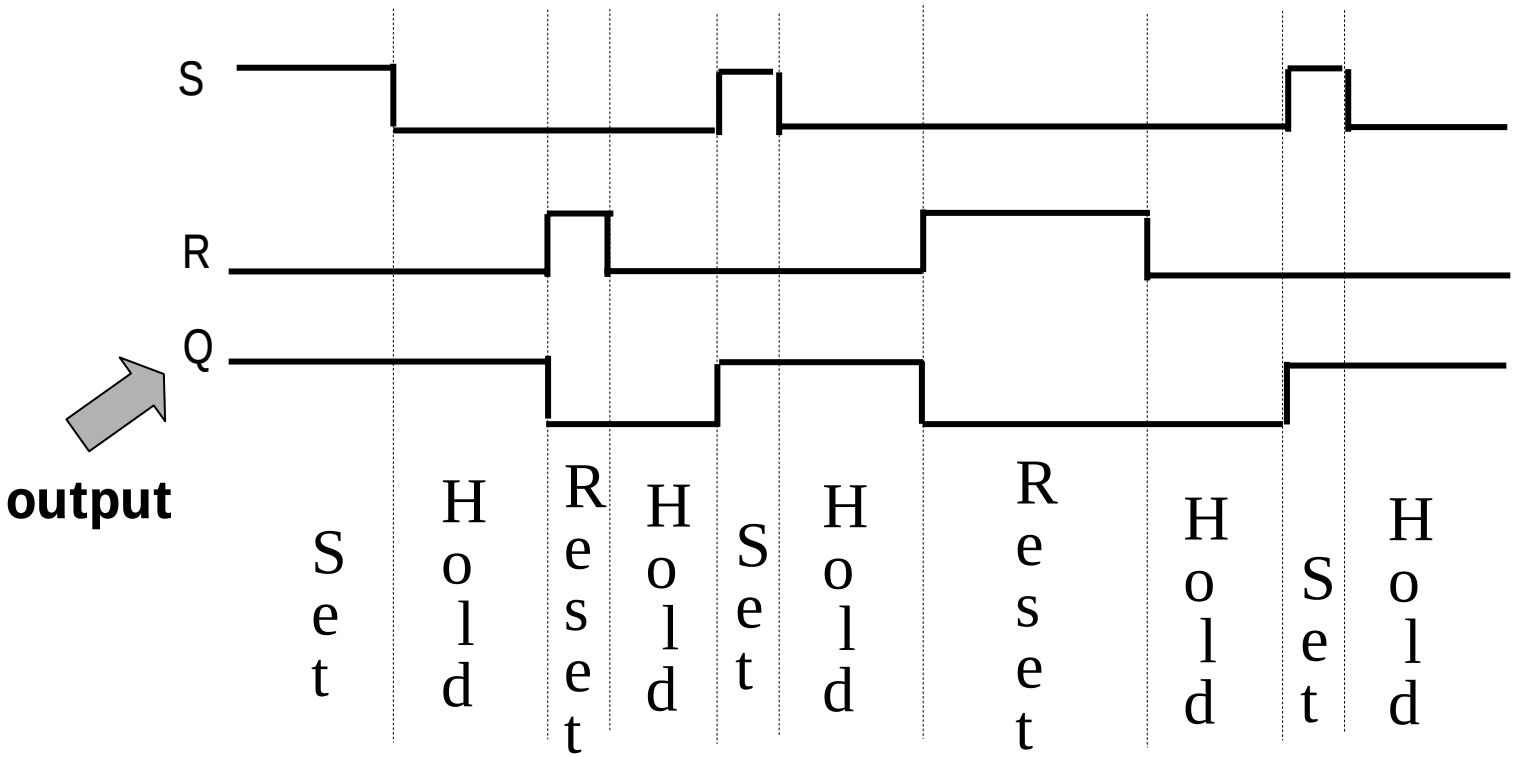


Simbol dari SR-FF

Timing Diagram sebuah SR-FF

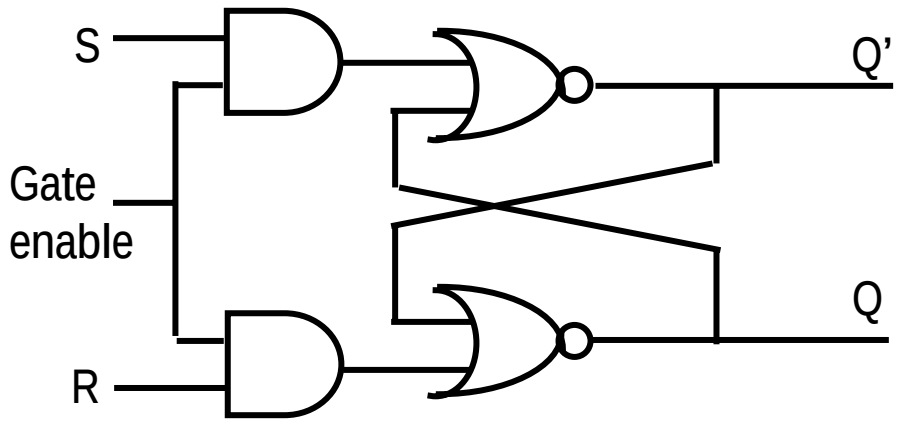
Diketahui :

timing diagram dari input S dan R pada sebuah SR-FF adalah seperti di bawah. Gambarkan timing diagram outputnya.



Gated SR-FF

- Rangkaian SR-FF yang diberi input tambahan : Gate
- Gate berfungsi mengontrol output dari SR-FF
- Gate/Clock merupakan rangkaian sinyal kontinyu
- Merupakan SR-FF sinkron (karena nilai output berubah sesuai dengan peng-aktifan input gate-nya).

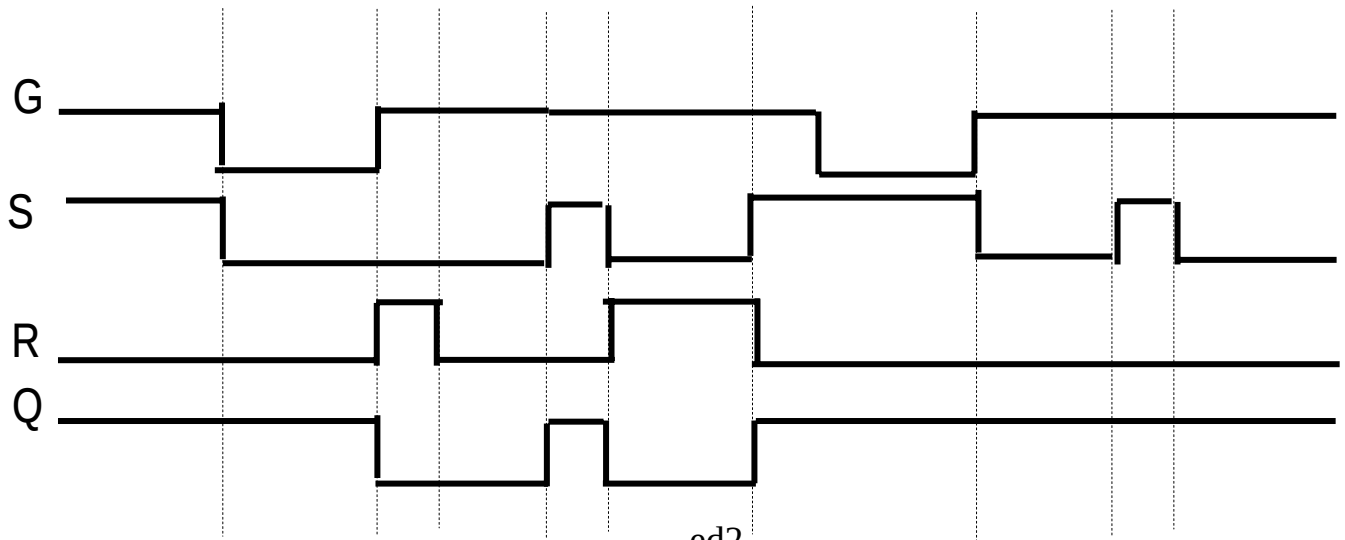


Gated SR-FF

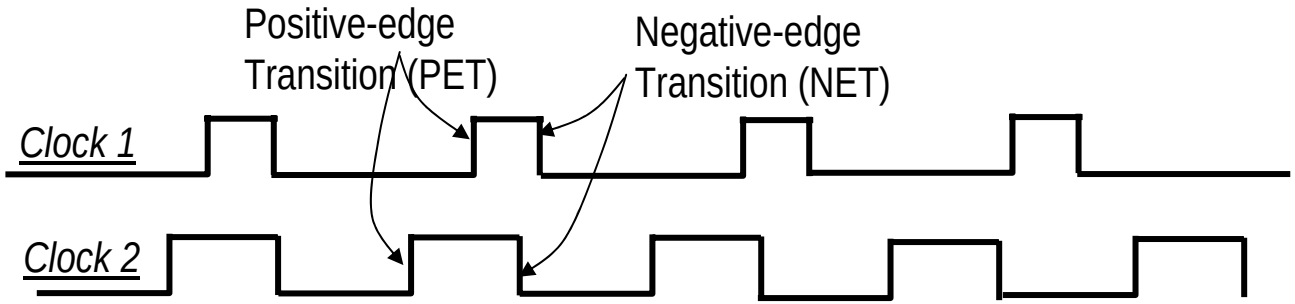
G	S	R	Q	Q'	COMMENT	
0	0	0	Q	Q'	Hold	Gate disable
0	0	1	Q	Q'	Hold	
0	1	0	Q	Q'	Hold	
0	1	1	Q	Q'	Hold	
1	0	0	Q	Q'	Hold	Gate enable
1	0	1	0	1	Reset	
1	1	0	1	0	Set	
1	1	1	0	0	Unused	

Tabel Fungsi dari Gated SR-FF

Timing Diagram Gated SR-FF



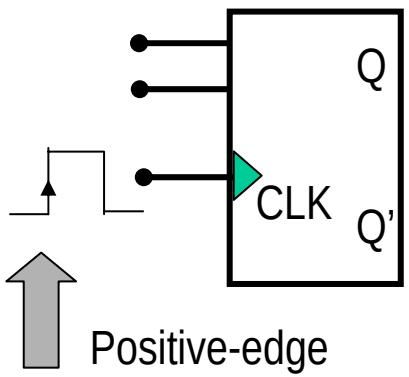
Sinyal Clock

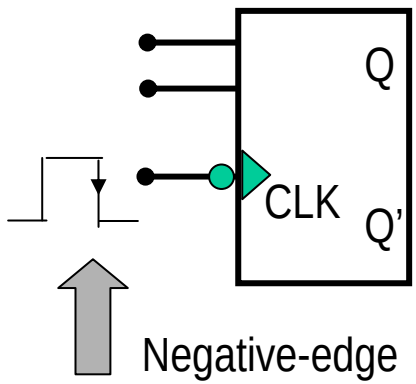


Positive-edge transition : saat clock berpindah dari 0 ke 1

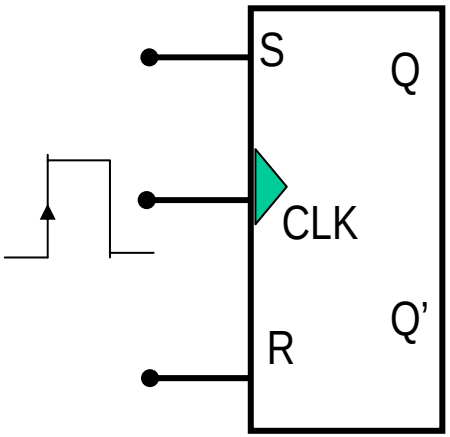
Negative-edge transition : saat clock berpindah dari 1 ke 0

Flip-Flop ber clock



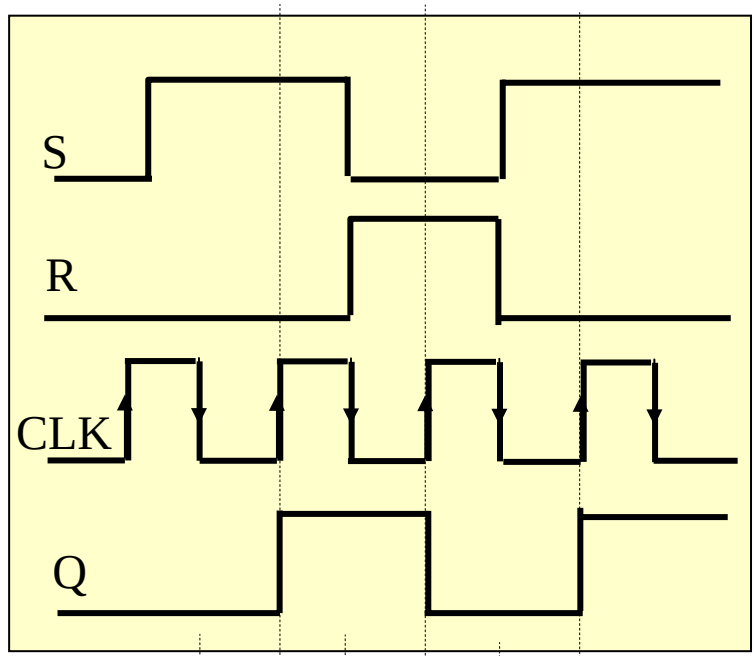


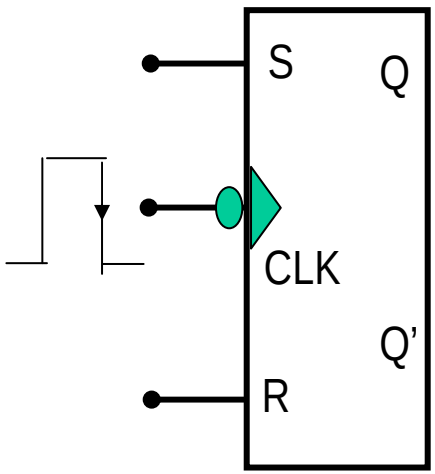
Clocked SR-FF



S	R	CLK	OUT
0	0	↑	Hold
0	1	↑	0
1	0	↑	1
1	1	↑	unused

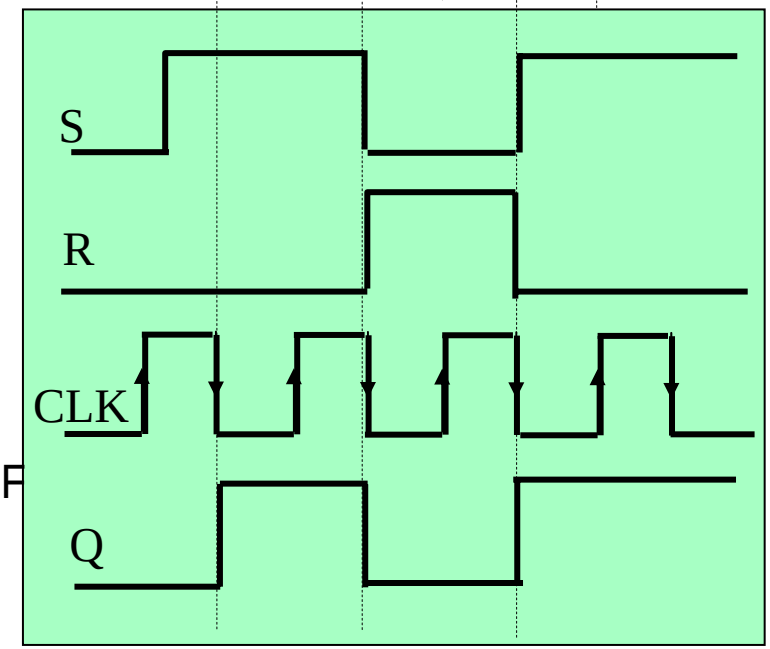
Positive-edge triggered SR-FF





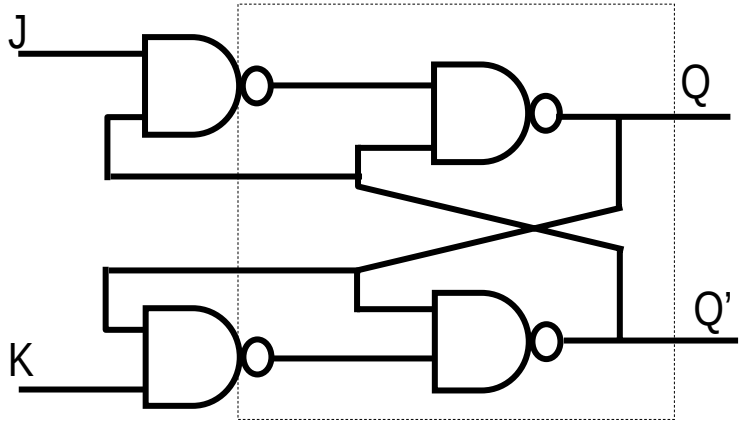
S	R	CLK	OUT
0	0	↓	Hold
0	1	↓	0
1	0	↓	1
1	1	↓	unused

Negative-edge triggered SR-FF

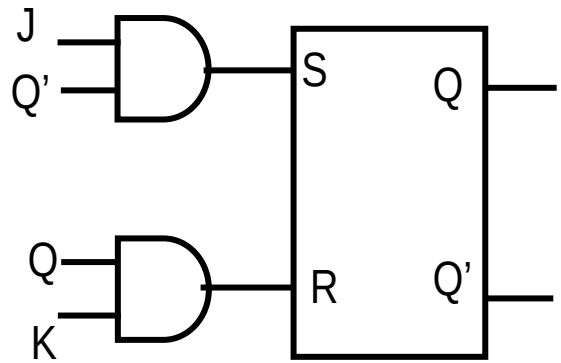


JK-FLIP-FLOP

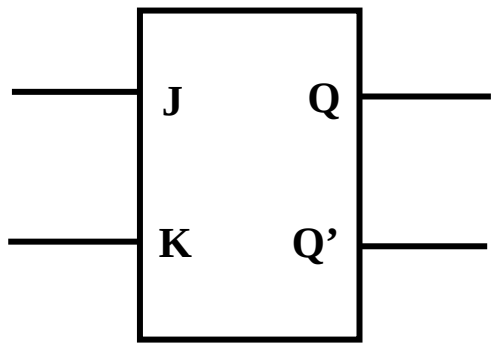
RANGKAIAN DASAR JK-FF



atau



Simbol dari JK-FF



Tabel State dari JK-FF

PRESENT INPUT		PRESENT OUTPUT	NEXT OUTPUT	Comment
J (t)	K (t)	Q (t)	Q (t+Δ)	
0	0	0	0	Hold
0	0	1	1	
0	1	0	0	Set
0	1	1	0	
1	0	0	1	Reset
1	0	1	1	
1	1	0	1	Toggle
1	1	1	0	

Tabel Eksitasi dari JK-FF

PRESENT OUTPUT	NEXT OUTPUT	NILAI EKSITASI	
Q (t)	Q (t+Δ)	J (t)	K (t)
0	0	0	d
0	1	1	d
1	0	d	1
1	1	d	0

DIketahui → Persamaan Next State SR-FF

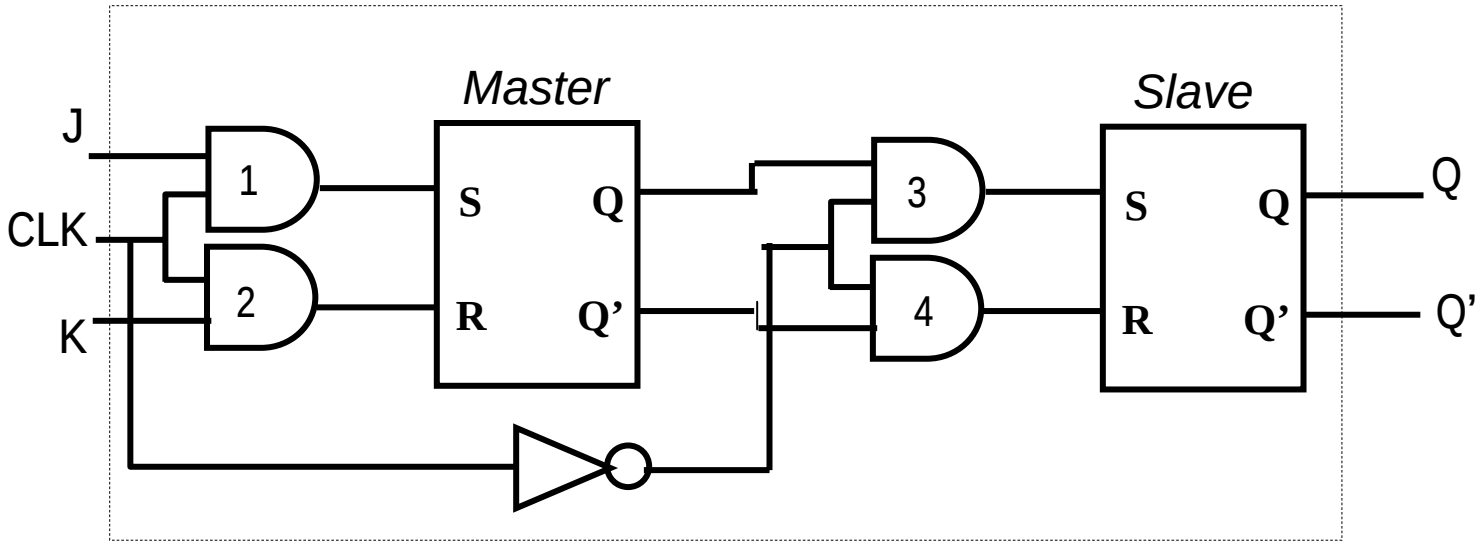
$$Q(t+\Delta) = S(t) + \overline{R}(t)Q(t)$$

Jika : $S(t) = J(t)\overline{Q}(t)$ dan $R(t) = K(t)Q(t)$ maka

Persamaan Next State JK-FF

$$Q(t+\Delta) = J(t)\overline{Q}(t) + \overline{K}(t)Q(t)$$

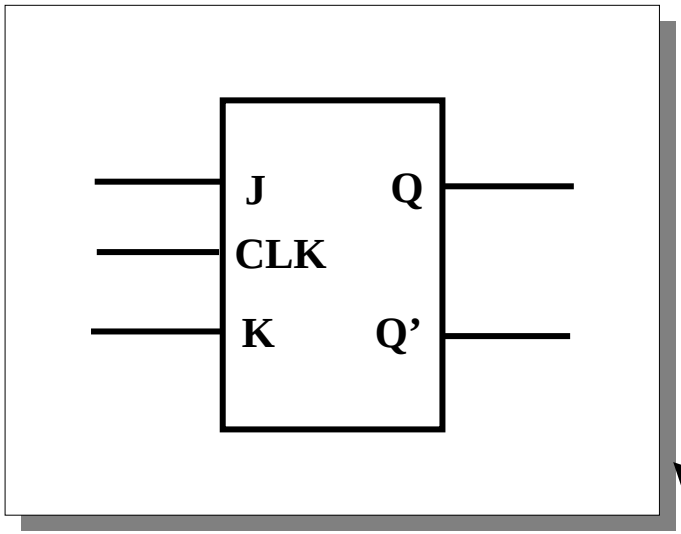
MASTER-SLAVE JK-FF



Rangkaian Ekivalen MS JK-FF

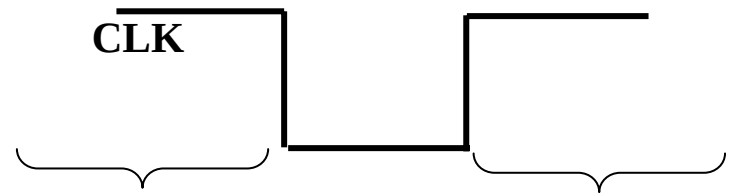
If CLK=1, gate 1 & 2 enable ⇨ Master ON
gate 3 & 4 disable ⇨ Slave OFF ⇨ input enable, output disable

If CLK=0, gate 1 & 2 disable ⇨ Master OFF
gate 3 & 4 enable ⇨ Slave ON ⇨ input disable, output enable



Simbol dari MS JK-FF

Timing diagram Clock



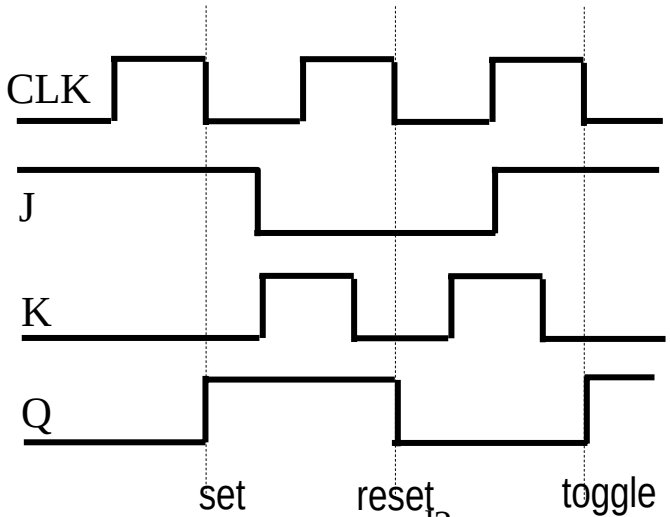
Gate 1 & 2 enable;
master loaded

Gate 1 & 2 disable;
Gate 3 & 4 enable;
slave loaded from master

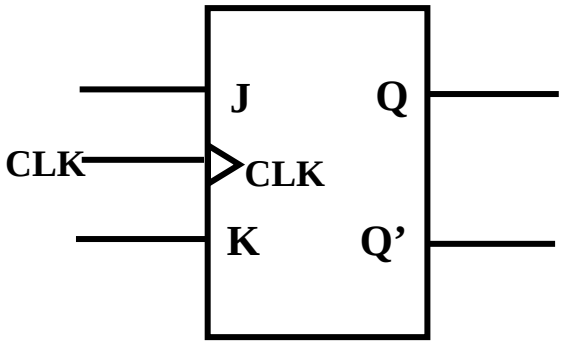
Cycle repeats

Positive-pulse triggered JK-FF

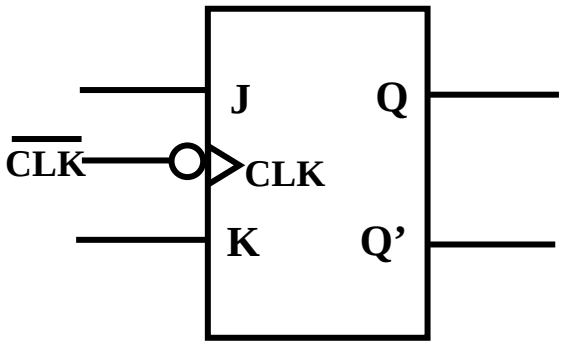
Timing diagram



Edge-triggered JK-FF



(a)

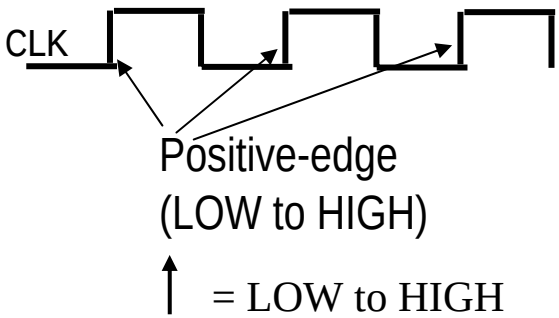


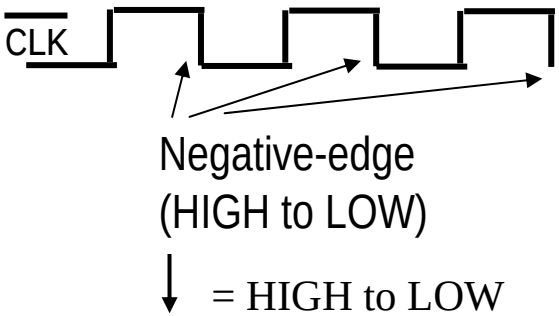
(b)

Simbol dari :

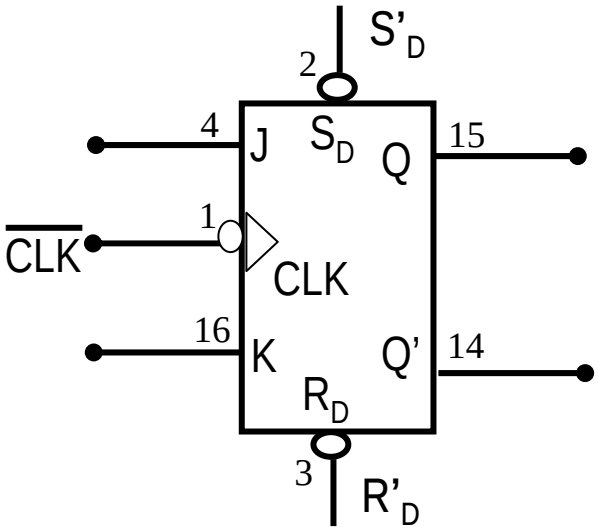
a) Positive-edge triggered JK-FF

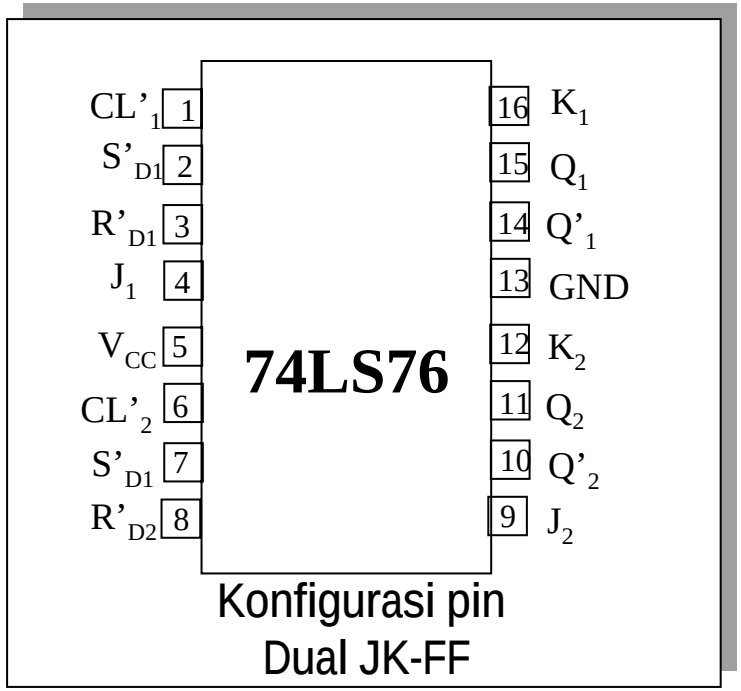
b) Negative-edge triggered JK-FF





JK-FF dengan input-input ASINKRON

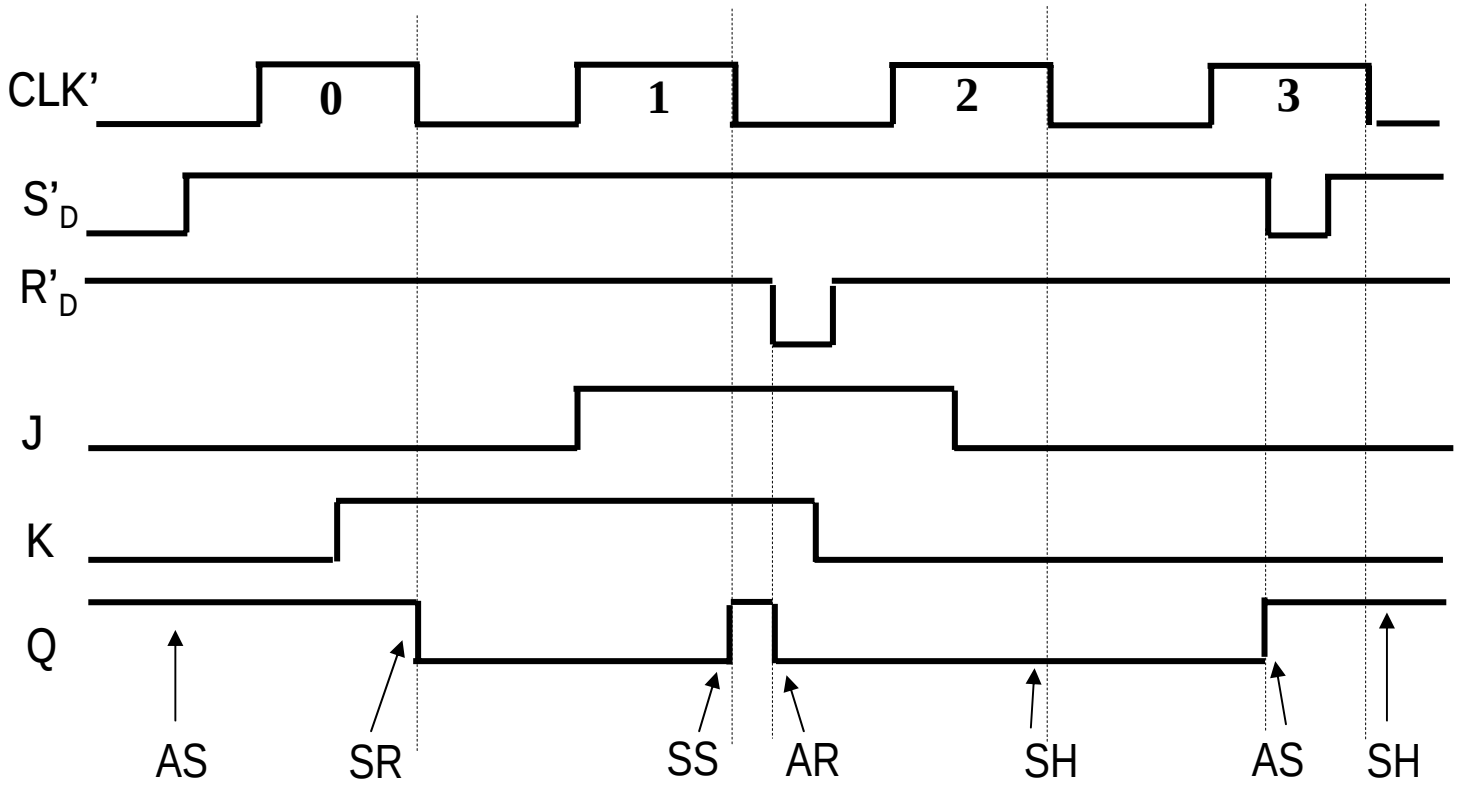




Konfigurasi pin
Dual JK-FF

OPERATING MODE	INPUT					OUTPUT
	S'_D	R'_D	CLK'	J	K	Q
Asynchronous Set	L	H	X	X	X	H
Asynchronous Reset	H	L	X	X	X	L
Synchronous Hold	H	H	↓	l	l	q
Synchronous Set	H	H	↓	h	l	H
Synchronous Reset	H	H	↓	l	h	L
Synchronous Toggle	H	H	↓	h	h	q'

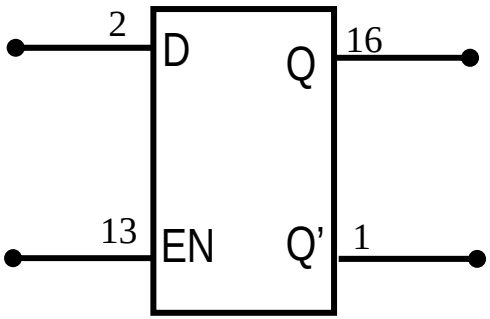
Timing diagram dari 74LS76 negative-edge triggered JK-FF



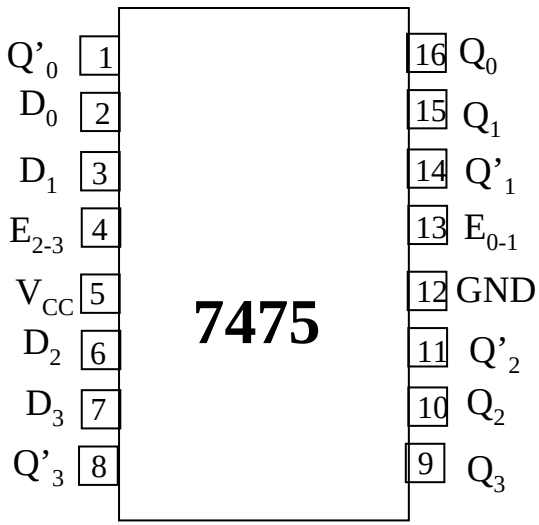
D-FLIP-FLOP

D-FF * = Data / delay Flip-flop

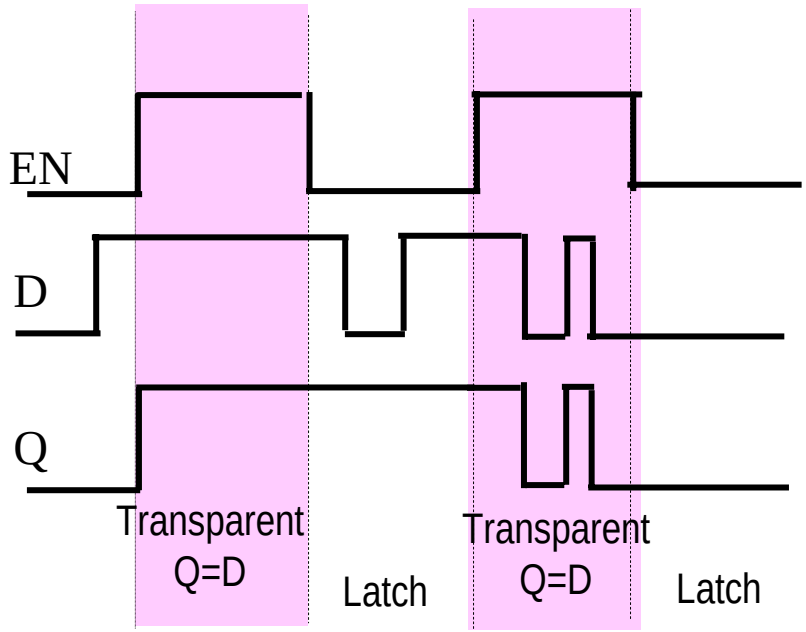
D-Latch (7475)



EN	D	Q	Comment
0	X	Q	Hold
1	0	0	Data '0'
1	1	1	Data '1'

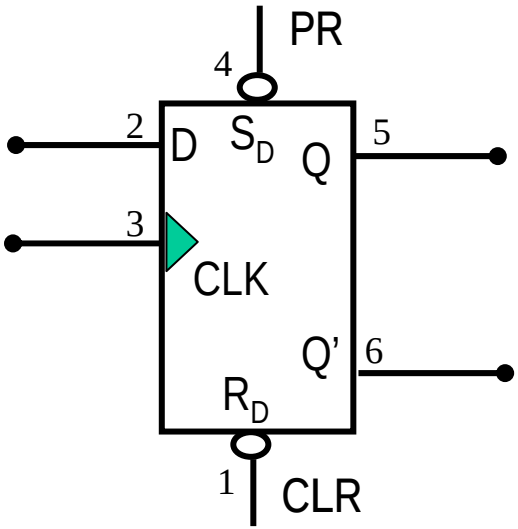


Konfigurasi pin dari Quad bistable D latch 7475



Timing Diagram dari D latch 7475

D-FF dengan INPUT ASINKRON

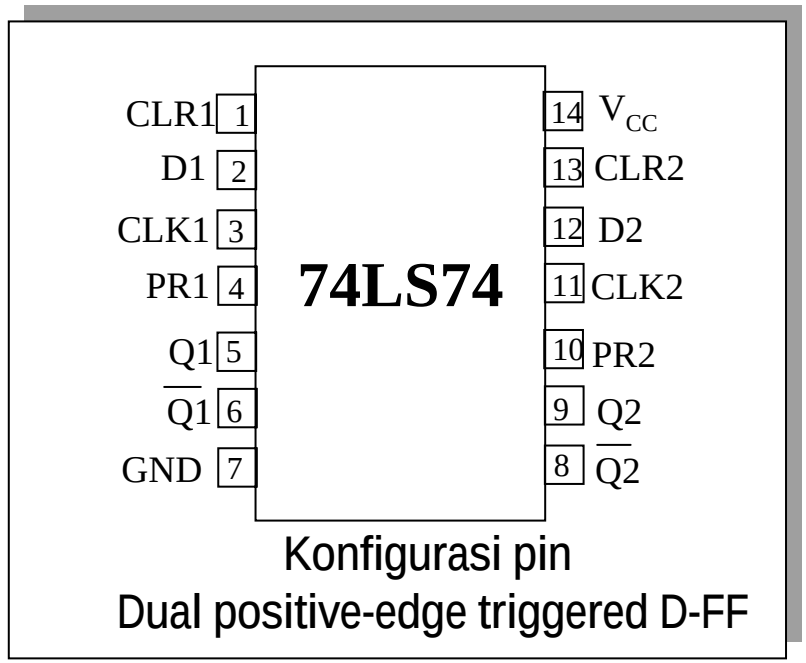


D-FF (7474)

	Input				Output
Operating Mode	S'_D	R'_D	CLK	D	Q
Asinkron Set	L	H	X	X	H
Asinkron Reset	H	L	X	X	L
Not used	L	L	X	X	H
Sinkron Set	H	H	↑	h	H
Sinkron Reset	H	H	↑	l	L

D,CLK = input *sinkron* (data,clock)

R'_D, S'_D = input *asinkron* (set,reset)

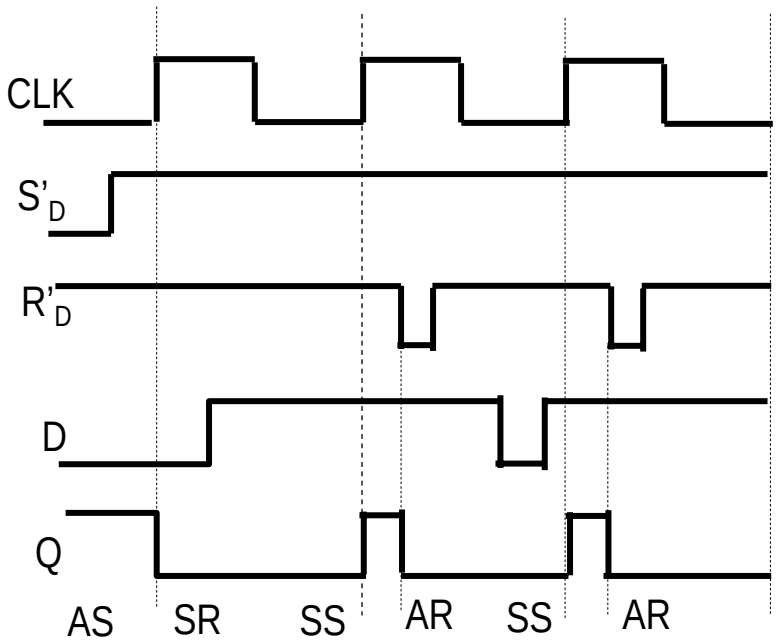


Konfigurasi pin
Dual positive-edge triggered D-FF

Tabel Eksitasi dari D-FF

PRESENT OUTPUT	NEXT OUTPUT	NILAI EKSITASI
Q(t)	Q(t+Δ)	D(t)
0	0	0
0	1	1
1	0	0
1	1	1

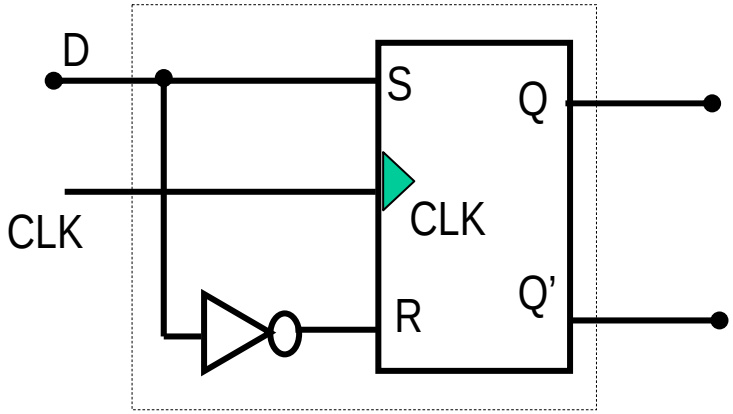
Timing Diagram



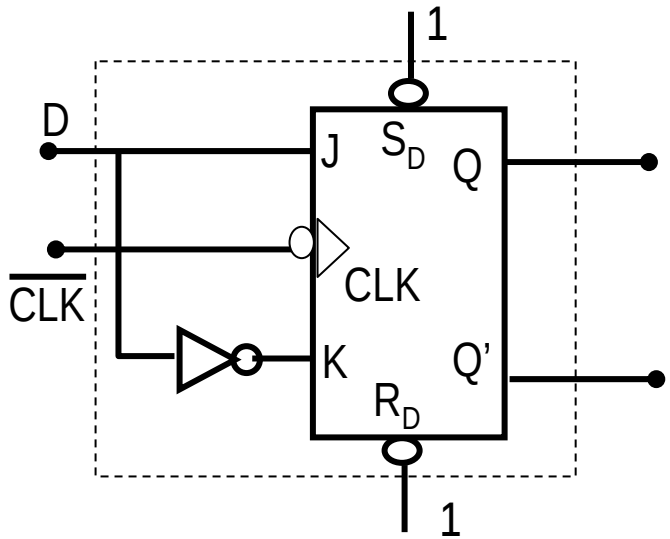
Persamaan Next State D-FF

$$Q(t+\Delta) = D(t)$$

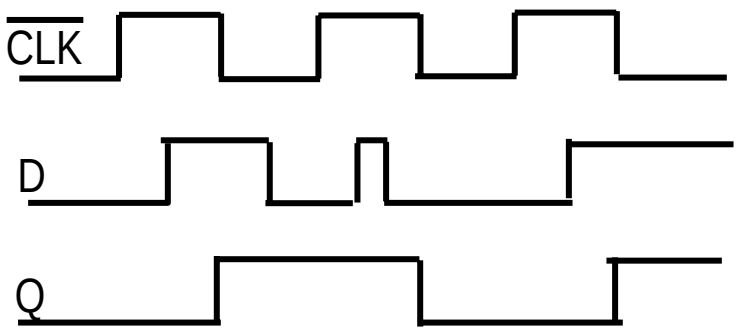
D-FF dari SR-FF



D-FF dari JK-FF



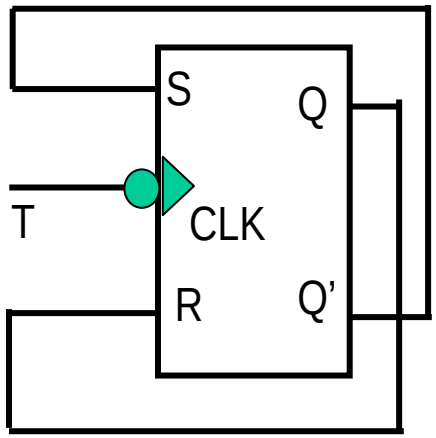
Timing diagram dari D-FF



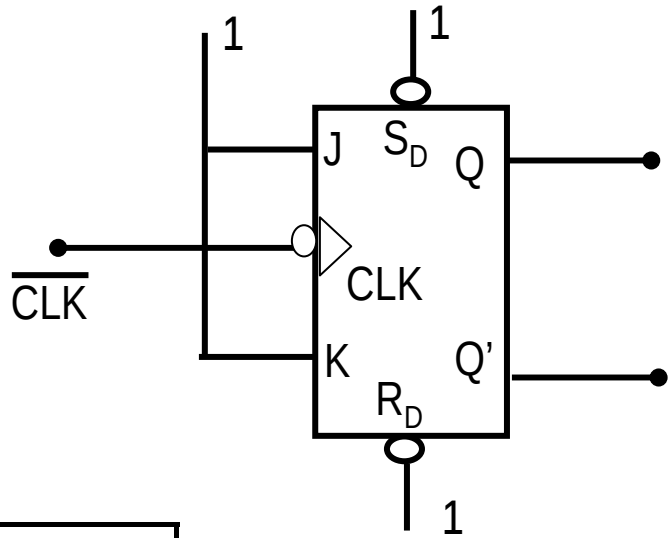
T-FLIP-FLOP

T-FF * = Toggle Flip-flop

T-FF dari SR-FF



T-FF dari JK-FF



T	Q	Comment
0	Q'	Toggle
1	Q	Hold

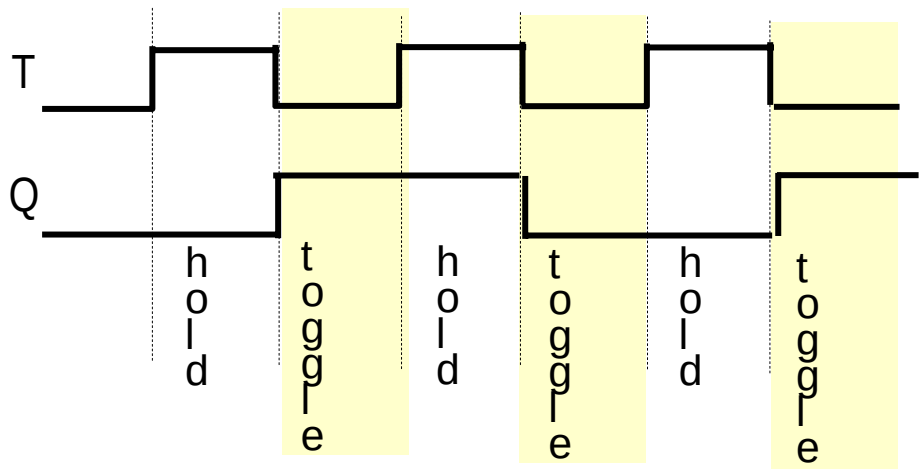
Tabel Eksitasi dari T-FF

PRESENT OUTPUT	NEXT OUTPUT	NILAI EKSITASI
Q(t)	Q(t+Δ)	T(t)
0	0	1
0	1	0
1	0	0
1	1	1

Persamaan Next State T-FF

$$Q(t+\Delta) = \overline{Q}(t)$$

Timing Diagram dari T-FF :



Analisa rangkaian

Prosedur meng-analisa rangkaian dengan Flip-flop

a. Tentukan persamaan logika kombinasional untuk input-input Flip-flopnya :
   input S dan R untuk SR-FF, input J dan K untuk JK-FF,
   input D untuk D-FF dan input T untuk T-FF

b. Untuk SR-FF →Tentukan apakah S.R = 0
   Catatan : Jika S.R ≠ 0, prosedur harus dihentikan.

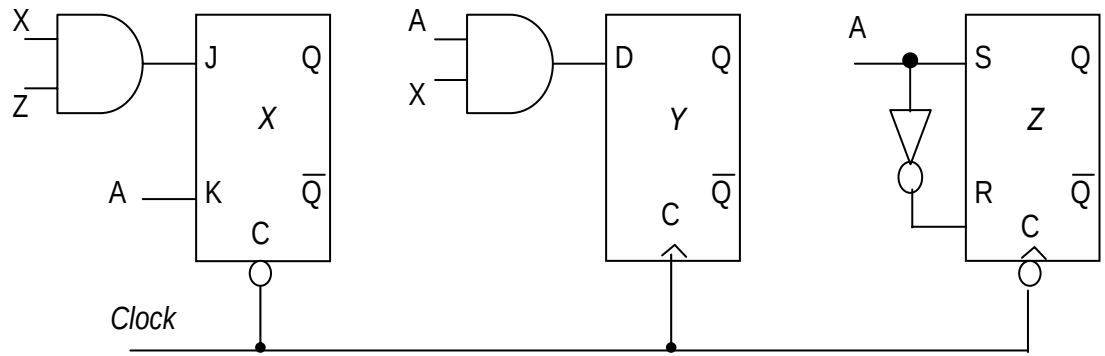
c. Cari persamaan Next State dari Flip-flop yang dicari :
   SR-FF → $Q(t+\Delta) = S(t) + \overline{R}(t)Q(t)$
   JK-FF → $Q(t+\Delta) = J(t)\overline{Q}(t) + \overline{K}(t)Q(t)$
   D-FF → $Q(t+\Delta) = D(t)$
   T-FF → $Q(t+\Delta) = \overline{Q}(t)$

d. Buat Tabel PS/NS – nya

e. Buat State Diagram-nya (jika perlu)

Contoh :

Carilah Tabel PS/NS dan State Diagram untuk rangkaian berikut ini :



Jawab :

Persamaan next state :

JK-FF

$$J(t) = X(t)Z(t)$$

$$K(t) = A(t)$$

$$X(t+\Delta) = J(t)\overline{X}(t) + \overline{K}(t)X(t)$$

$$= X(t)Z(t)\overline{X}(t) + \overline{A}(t)X(t) = \overline{A}(t)X(t)$$

D-FF

$$Y(t+\Delta) = D(t) = A(t)X(t)$$

SR-FF

$$S(t) = A(t) \qquad R(t) = \overline{A}(t)$$

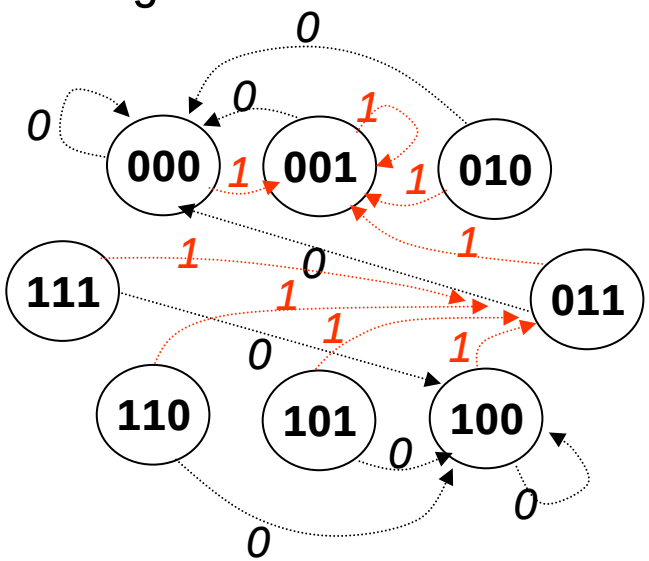
$$S(t).R(t) = A(t).\overline{A}(t) = 0$$

$$Z(t+\Delta) = S(t) + \overline{R}(t)Z(t)$$

$$= A(t) + \overline{\overline{A}}(t)Z(t)$$

$$= A(t)\left[1 + Z(t)\right] = A(t)$$

State Diagram



Tabel PS/NS

A(t)	X(t)	Y(t)	Z(t)	X(t+Δ)	Y(t+Δ)	Z(t+Δ)
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	0	1	0	0	0	0
0	0	1	1	0	0	0
0	1	0	0	1	0	0
0	1	0	1	1	0	0
0	1	1	0	1	0	0
0	1	1	1	1	0	0
1	0	0	0	0	0	1
1	0	0	1	0	0	1
1	0	1	0	0	0	1
1	0	1	1	0	0	1
1	1	0	0	0	1	1
1	1	0	1	0	1	1
1	1	1	0	0	1	1
1	1	1	1	0	1	1

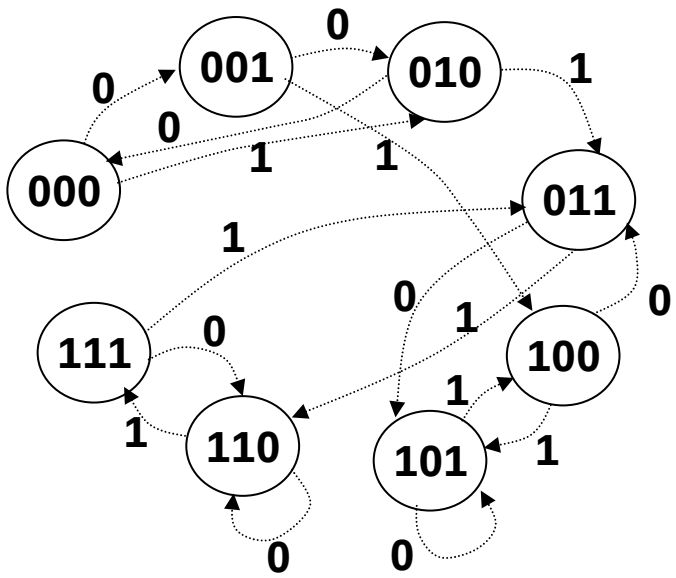
Disain/Sintesa rangkaian

Prosedur mendisain rangkaian dengan Flip-flop

1. Dengan menggunakan persamaan next state atau State Diagram yang diketahui, buatlah tabel present state/next state untuk rangkaian yang akan dibangun.
2. Tambahkan kolom pasangan eksitasi dari masing-masing Flip-flop yang akan digunakan.
3. Dengan menggunakan K-Map, carilah persamaan logika dari nilai eksitasi yang didapat
4. Buat rangkaian sesuai dengan persamaan yang didapat.

Contoh :

Diketahui sebuah State Diagram dari rangkaian sekuensial dengan D-FF seperti dibawah ini. Gambarkan bentuk rangkaiannya.



Jawab :

Tabel PS/NS

A	X	Y	Z	Xn	Yn	Zn
0	0	0	0	0	0	1
0	0	0	1	0	1	0
0	0	1	0	0	0	0
0	0	1	1	1	0	1
0	1	0	0	0	1	1
0	1	0	1	1	0	1
0	1	1	0	1	1	0
0	1	1	1	1	1	0
1	0	0	0	0	1	0
1	0	0	1	1	0	0
1	0	1	0	0	1	1
1	0	1	1	1	1	0
1	1	0	0	1	0	1
1	1	0	1	1	0	0
1	1	1	0	1	1	1
1	1	1	1	0	1	1

Tabel PS/NS dan Nilai Eksitasi dari D-FF

PI	PO			NO			Eksitasi		
A	X	Y	Z	Xn	Yn	Zn	Dx	Dy	Dz
0	0	0	0	0	0	1	0	0	1
0	0	0	1	0	1	0	0	1	0
0	0	1	0	0	0	0	0	0	0
0	0	1	1	1	0	1	1	0	1
0	1	0	0	0	1	1	0	1	1
0	1	0	1	1	0	1	1	0	1
0	1	1	0	1	1	0	1	1	0
0	1	1	1	1	1	0	1	1	0
1	0	0	0	0	1	0	0	1	0
1	0	0	1	1	0	0	1	0	0
1	0	1	0	0	1	1	0	1	1
1	0	1	1	1	1	0	1	1	0
1	1	0	0	1	0	1	1	0	1
1	1	0	1	1	0	0	1	0	0
1	1	1	0	1	1	1	1	1	1
1	1	1	1	0	1	1	0	1	1

AX \ YZ	00	01	11	10
00	1	0	1	0
01	1	1	0	0
11	1	0	1	1
10	0	0	0	1

$$Dz = \bar{A}\bar{Y}\bar{Z} + X\bar{Y}\bar{Z} + \bar{A}X\bar{Y} + AXY + AY\bar{Z} + \bar{A}\bar{X}YZ$$

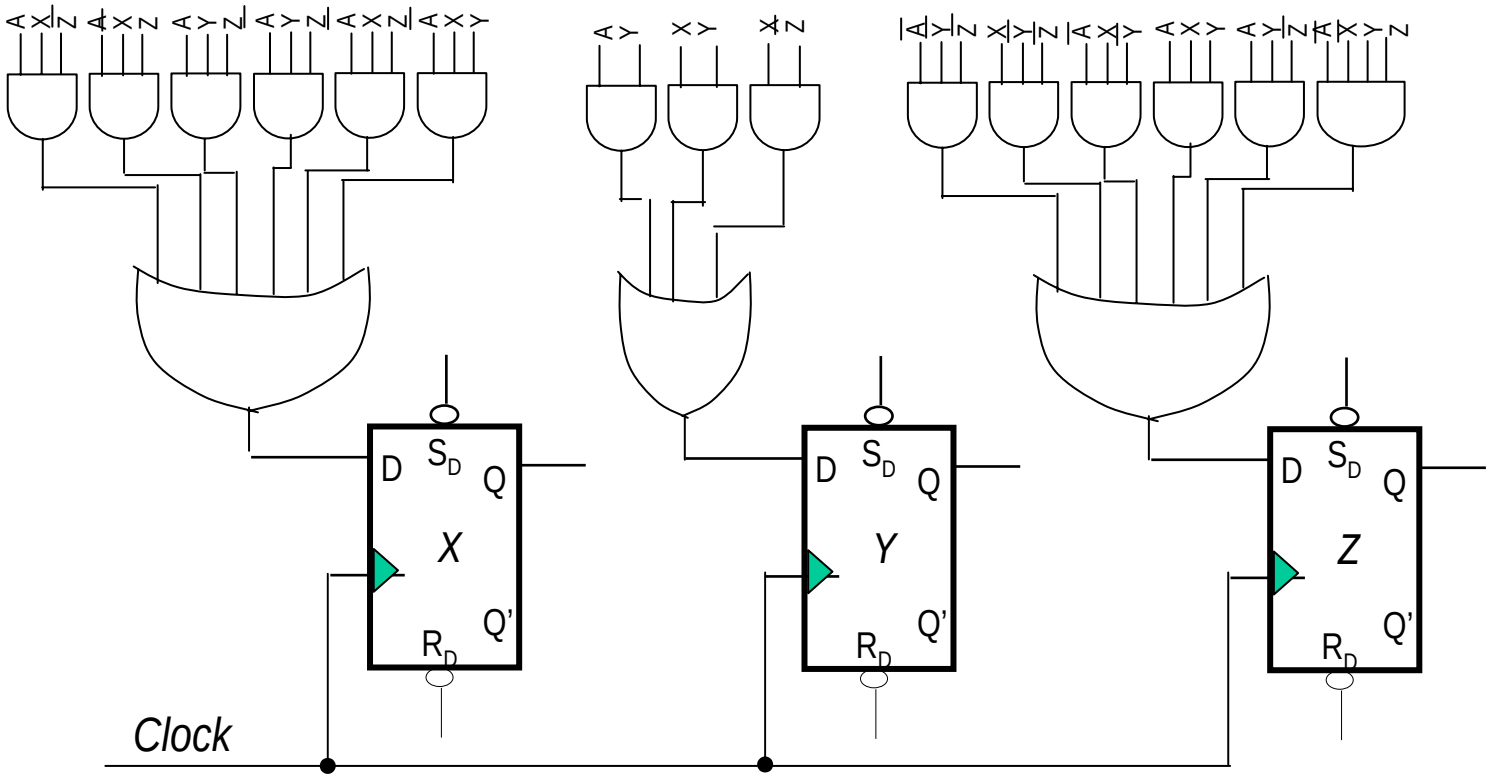
AX \ YZ	00	01	11	10
00	0	0	1	0
01	0	1	1	1
11	1	1	0	1
10	0	1	1	0

$$Dx = AX\bar{Z} + A\bar{X}Z + AYZ + \bar{A}YZ + \bar{A}XZ + \bar{A}XY$$

AX \ YZ	00	01	11	10
00	0	1	0	0
01	1	0	1	1
11	1	0	1	1
10	0	0	1	1

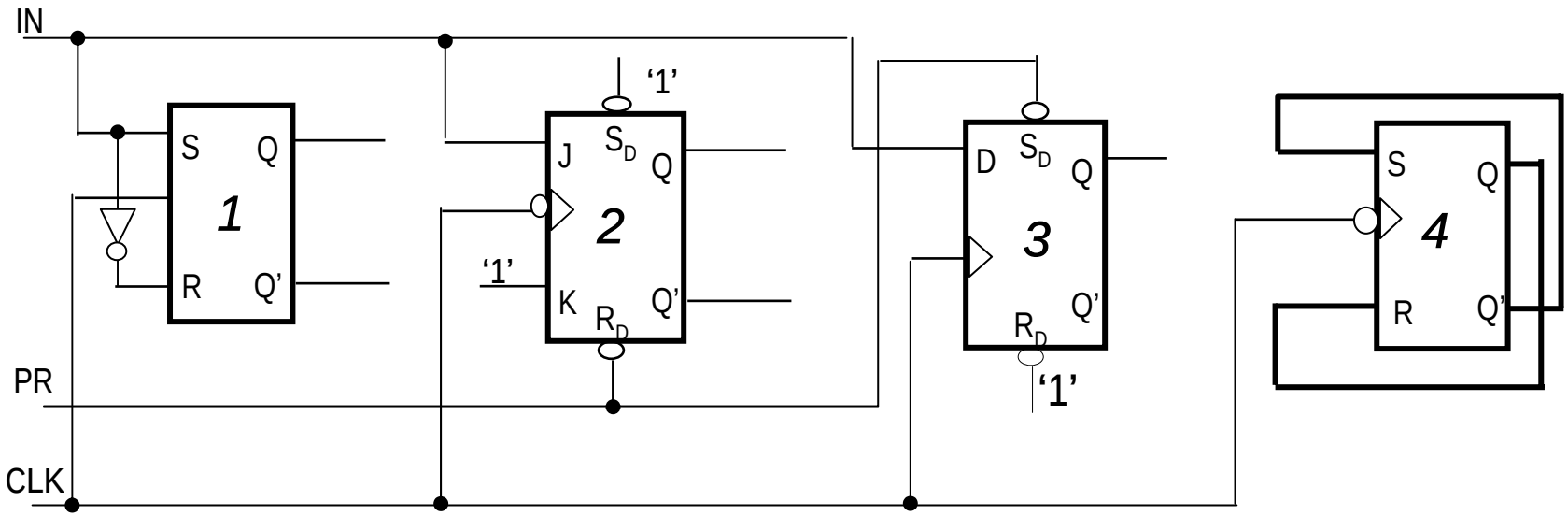
$$Dy = AY + XY + X\bar{Z}$$

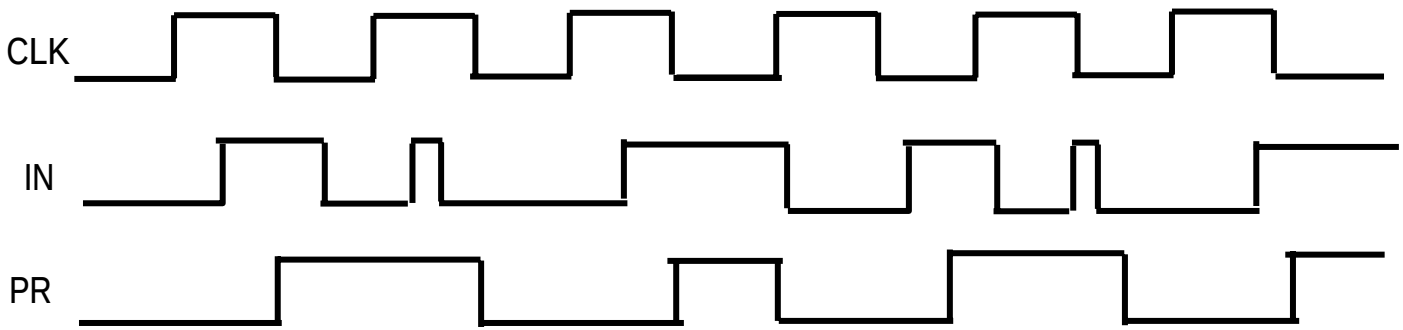
Gambar rangkaian



Soal Latihan

1. Gambarkan bentuk gelombang output untuk beberapa jenis Flip-flop di bawah ini, jika diketahui bentuk gelombang inputnya adalah sebagai berikut :





Q_1, Q_2, Q_3, Q_4 ....??